

Operating System

Chapter 2. OS Overview



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Class Information



❑ **Lecturer**

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❑ **Time**

- Thu 10:30am – 12:15pm

❑ **Place**

- New Engineering Building 106

❑ **Textbook**

- “[Operating Systems: Internals and Design Principles](#)”, William Stallings, Pearson, 9th Edition, 2017.

❑ **References**

- “[Computer Systems: A Programmer's Perspective](#)”, Randal E. Bryant and David O'Hallaron, Prentice Hall, 3rd Edition, 2016.

❑ **Class homepage**

- <http://kulms.korea.ac.kr>, <http://it.korea.ac.kr> : slides, announcements



□ ***Course overview***

- 1. OS Overview
- 2. Process
- 3. Thread
- 4. Mutual Exclusion and Synchronization
- 5. Deadlock and Starvation
- 6. Memory Management
- 7. Virtual Memory
- 8. Uniprocessor Scheduling
- 9. Multiprocessor and Realtime Scheduling
- 10. IO
- 11. File Management
- 12. Virtual Machine

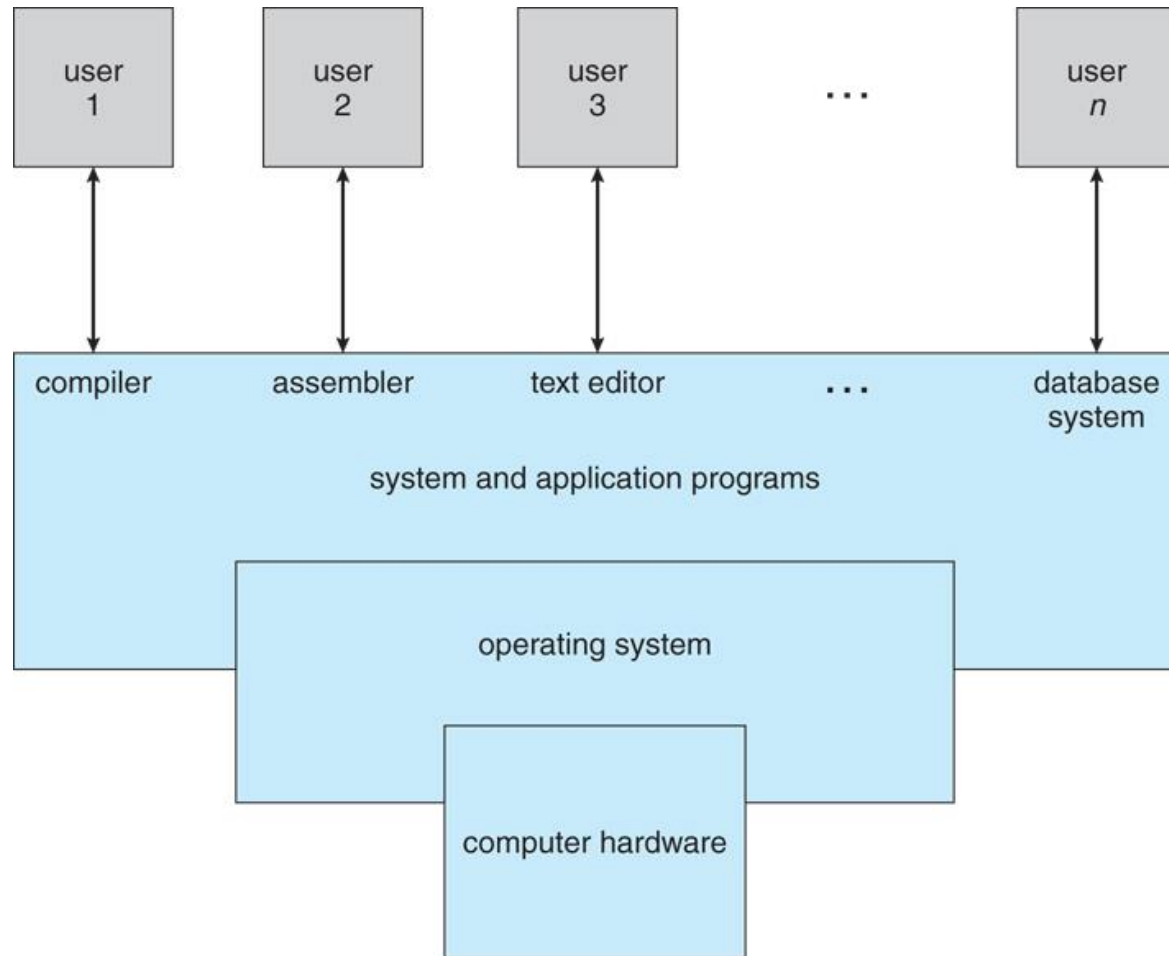
Class Information



□ ***Evaluation***

- Midterm : 35%
- Final: 35%
- Homework and Projects: 30%
- Class participation: extra 5%
 - Attendance: no shows of more than 2 will get -5%
 - Bonus points

Abstract view of a computer system

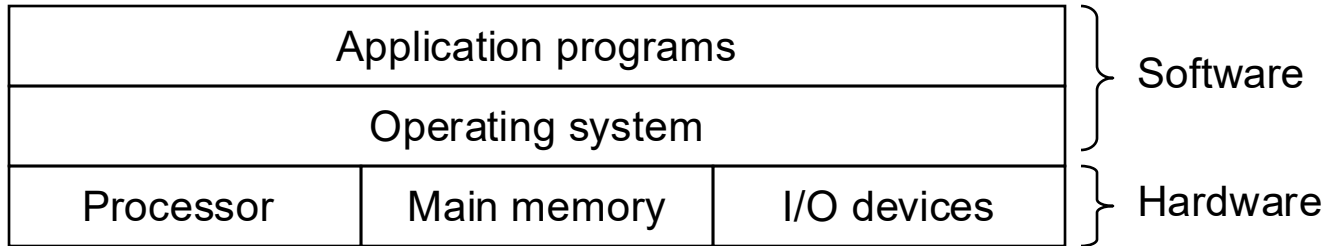


Operating system controls the hardware and coordinates its use among various application programs for various users

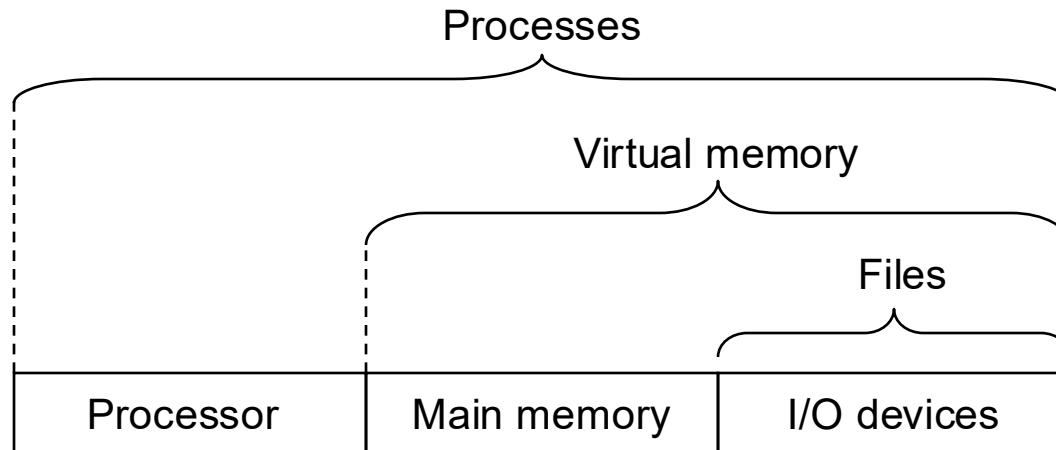


❑ ***Operating system***

- A layer of software between application programs and the hardware
- Three purposes
 - **Easy of use:** Provide applications with simple and uniform interface to manipulate complicated and often widely different low-level hardware devices
 - **Sharing:** Share the hardware resource from multiple processes/users and increase the **resource utilization**
 - **Protection:** Protect the hardware from misuse by runaway applications
 - ▼ Incorrect or malicious programs should not cause other programs to execute incorrectly
 - ▼ Smartphone and PC operating systems emphasize easy of use while server and mainframe OSs emphasize sharing, protection, and fairness
- Commercial operating systems usually provide not only **kernels** (*programs running at all times on the computer*) but also system **middleware, system programs**, and application programs.
 - Windows 76.5%, Mac OS 17.7%, Linux 1.7% (Mobile: Android 72%, IOS 27%)
- Use abstractions such as **processes, virtual memory, and files**



Layered view of a computer system



Abstractions provided by an OS

Operating System



- ❑ ***An OS is a program that provides convenient services for application programs by managing hardware resources and acts as an interface between applications and the computer hardware***

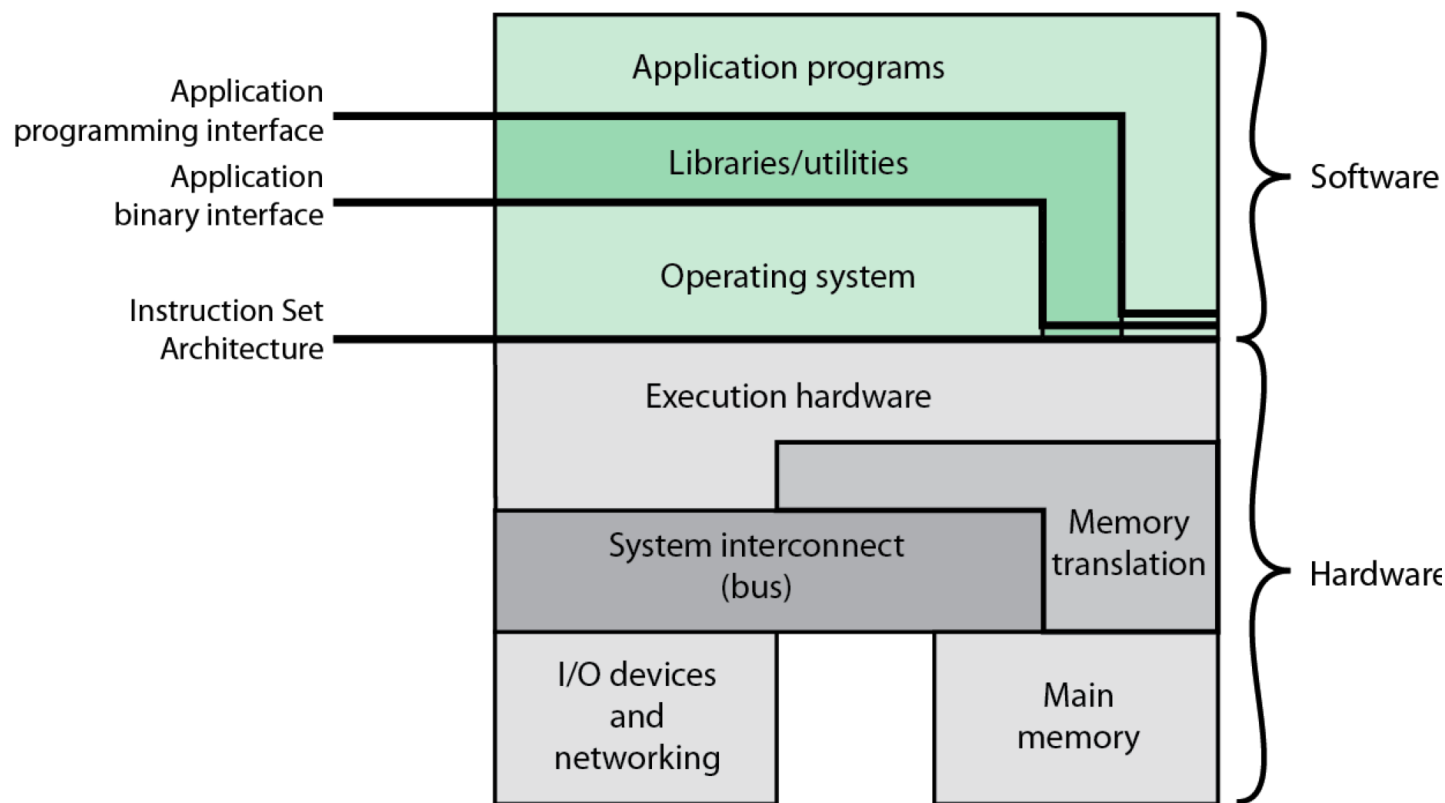


Figure 2.1 Computer Hardware and Software Infrastructure

Source: Pearson

Key Interfaces



❑ ***Instruction set architecture (ISA)***

- Define the interface between SW and HW
 - A set of machine language instructions
 - Both application programs and utilities can access the ISA directly

❑ ***Application binary interface (ABI)***

- Define the system call interface to OS

❑ ***Application programming interface (API)***

- Define the program call interface to system services. System calls are performed through library calls.
 - Enables application programs to be ported easily, through recompilation, to other systems that support the same API

Terminology



❑ **Microprocessor: a single chip processor**

- Intel i7, Pentium IV, AMD Ryzen, SPARC T5, ..

❑ ***ISA (Instruction Set Architecture)***

- Defines machine instructions and visible machine states (registers + memory)
- Examples
 - IA32(x86), IA64, ARM, MIPS, SPARC, PowerPC
 - x86 family: Intel Pentium, Pentium Pro, Pentium 4, AMD Athlon
 - x86-64 family, Intel i3, i5, i7, i9
 - IA64 family: Itanium, Itanium2, Itanium 9300/9500/9700
 - ARM family: ARM7, ARM11, ARM Cortex-A53

❑ ***Microarchitecture***

- Implementation: HW design and implementation according to the ISA
 - Pipelining, caches, branch prediction, buffers, out-of-order execution
 - 80386, 80486, Pentium, Pentium Pro, Pentium 4 are the 1st, 2nd, 3rd, 4th, 5th implementation (microarchitecture) of x86 ISA
- Invisible to programmers
 - From the programmer's perspective, both 80486 and Pentium 4 support the same ISA and show the same behavior for a machine instruction

Terminology



❑ *CISC (Complex Instruction Set Computer)*

- Each instruction is complex
 - Instructions of different sizes, many instruction formats, allow computations on memory data, ...
- A large number of instructions in ISA
- Architectures until mid 80's
 - Examples: x86, VAX

❑ *RISC (Reduced Instruction Set Computer)*

- Each instruction is simple
 - Fixed size instructions, only a few instruction formats
- A small number of instructions in ISA
- Load-store architectures
 - Computations are allowed only on registers
 - ▾ Data must be transferred to registers before computation
- Most architectures built since 80's
 - Examples: MIPS, ARM, PowerPC, Alpha, SPARC, IA64, PA-RISC, etc.

Terminology



□ *Word*

- Default data size for computation
 - Size of a GPR & ALU data path depends on the word size
 - ▼ GPR stands for general purpose (integer) registers
 - ▼ ALU stands for arithmetic and logic unit
- The word size determines if a processor is a 8b, 16b, 32b, or 64b processor

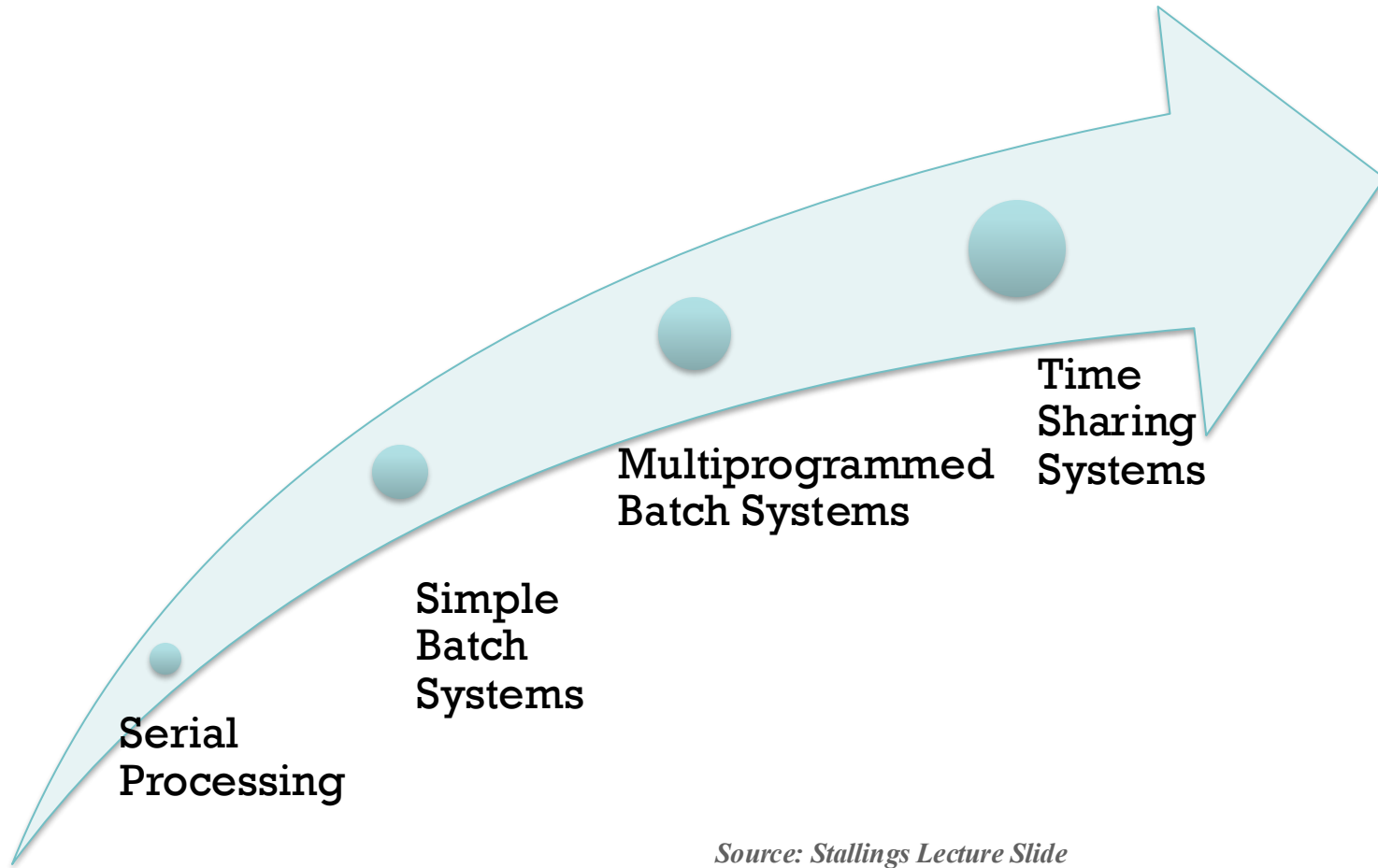
□ *Address (or pointer)*

- Points to a location in memory
- Each address points to a byte (*byte addressable*)
 - If you have a 32b address, you can address 2^{32} bytes = 4GB
 - If you have a 256MB memory, you need at least 28 bit address since $2^{28} = 256\text{MB}$

□ *Caches*

- Faster but smaller memory close to processor
 - Fast since they are built using SRAMs
 - Smaller since they are expensive
 - Usually inside CPU chip (located between processor core and main memory)

Evolution of Operating Systems





❑ ***Earliest computers***

- No operating system until mid 1950s
 - Programmers interacted directly with the computer hardware
- Computers ran from a console with display lights, toggle switches, jumper cables, some form of input device, and a printer

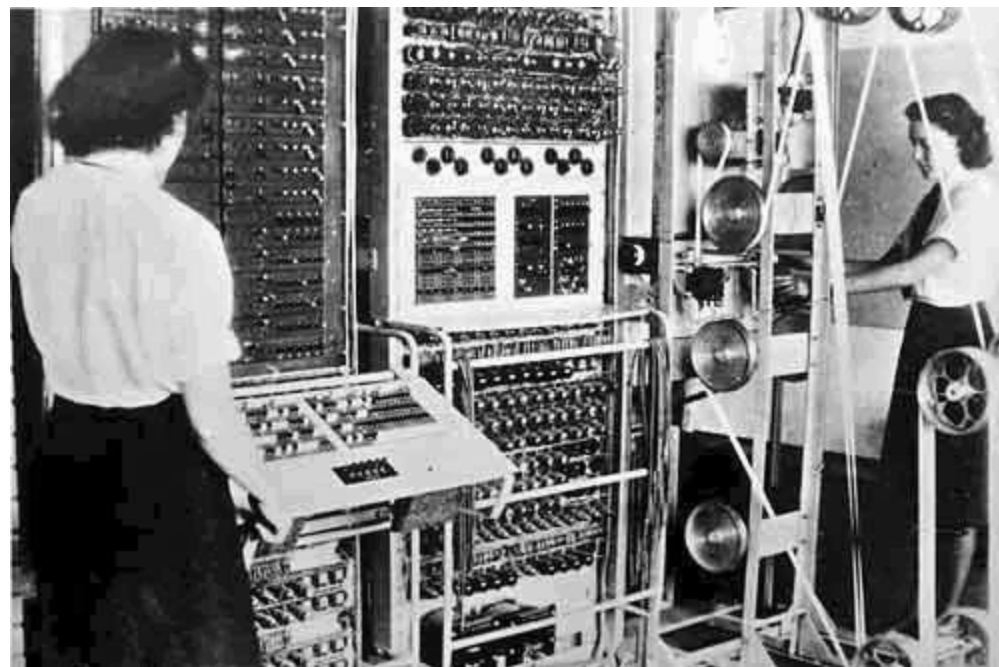
❑ ***Problems***

- Scheduling
 - Most installations used a *hardcopy sign-up sheet to reserve computer time*. However, time allocations could *run short or long*, resulting in *wasted time*
- Setup time
 - A considerable amount of time was spent just on setting up the program to run. Compile/link/load require mounting tapes, setting up card decks, etc.
- Early computers were very expensive
 - Important to maximize processor utilization

Alan Turing, Bombe and Colossus



Bletchley Park



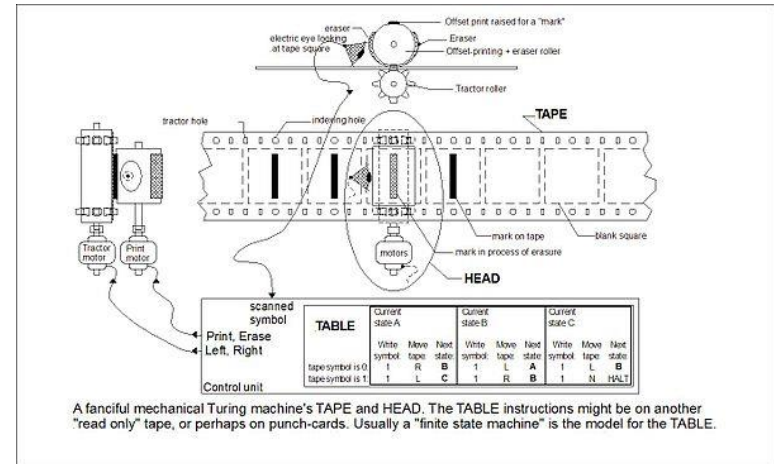
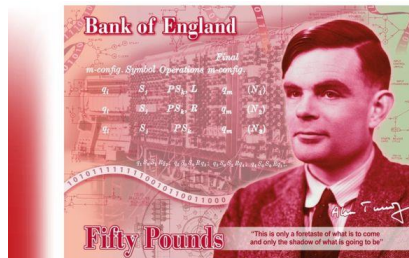
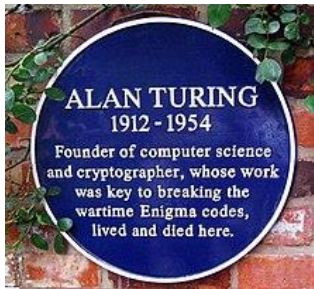
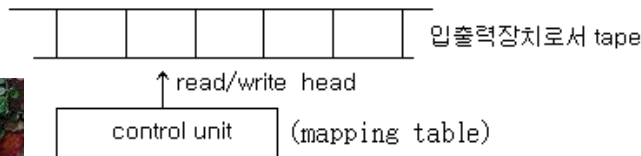
Source: Wikipedia

Alan Turing (1912 ~ 1954)



❑ **Turing Machine (1936)**

- A new mathematical computing model
 - An abstract machine that manipulates symbols on a tape according to a table of rules
- A symbol manipulating device that can simulate the logic of any computer
 - Consists of infinite linear tape, read/write head, and control unit
- Theoretical background to modern computers



A fanciful mechanical Turing machine's TAPE and HEAD. The TABLE instructions might be on another "read only" tape, or perhaps on punch-cards. Usually a "finite state machine" is the model for the TABLE.

❑ **Bombe (1940)**

- A electromechanical machine to decipher German's Enigma

❑ **Turing Test (1950)**

- A computer could be said to "think" if a human interrogator could not tell it apart, through conversation, from a human being

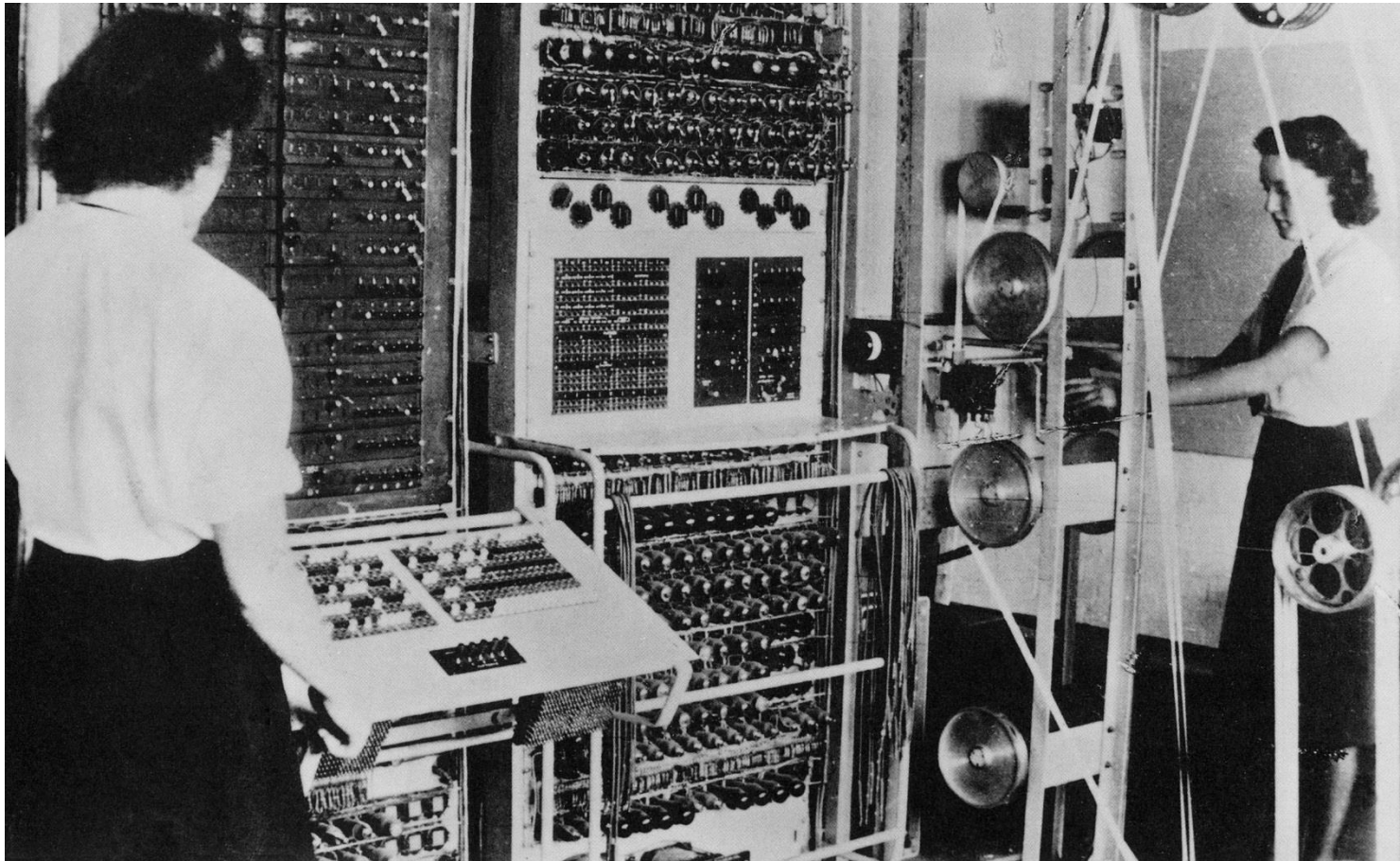
❑ **ACM Turing Award – the Novel prize in Computer Science**

Colossus



❑ *Colossus (1943)*

- The 1st programmable digital (electronic) computer built by Thomas Flowers in London

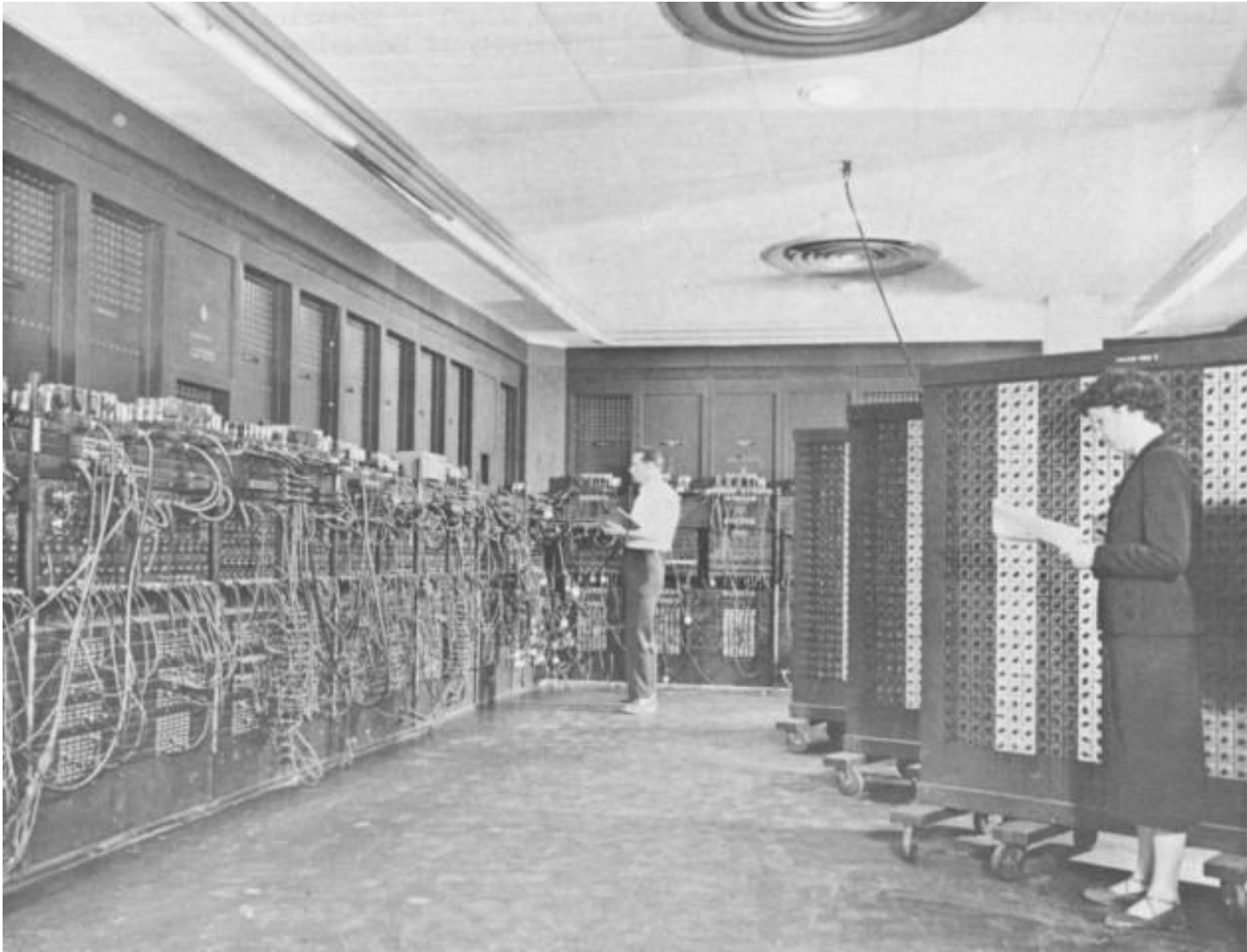




□ **ENIAC (Electronic Numerical Integrator And Computer)**

- Designed by John Mauchly and John Presper Eckert at University of Pennsylvania
- Funded by US BRL (Ballistic Research Lab) to develop range and trajectory tables for new weapons
 - Until then, BRL employee more than 200 people with desktop calculators to solve the necessary ballistics equations
 - The proposal accepted in 1943, the machine completed in 1946, and dismantled in 1955
- Used for H-bomb research
- Characteristics
 - 30 tons, 15000 square feet, 18000 vacuum tubes, 140 KW power dissipation
 - Decimal machine
 - ▼ 20 accumulators each holding 10-digit decimal number
 - ▼ Each digit is represented by a ring of 10 vacuum tubes
 - *Manually programmed by setting switches and plugging/unplugging cables*
 - 5,000 additions per second

ENIAC



Source: Wikipedia

The Von Neumann Machine & IAS

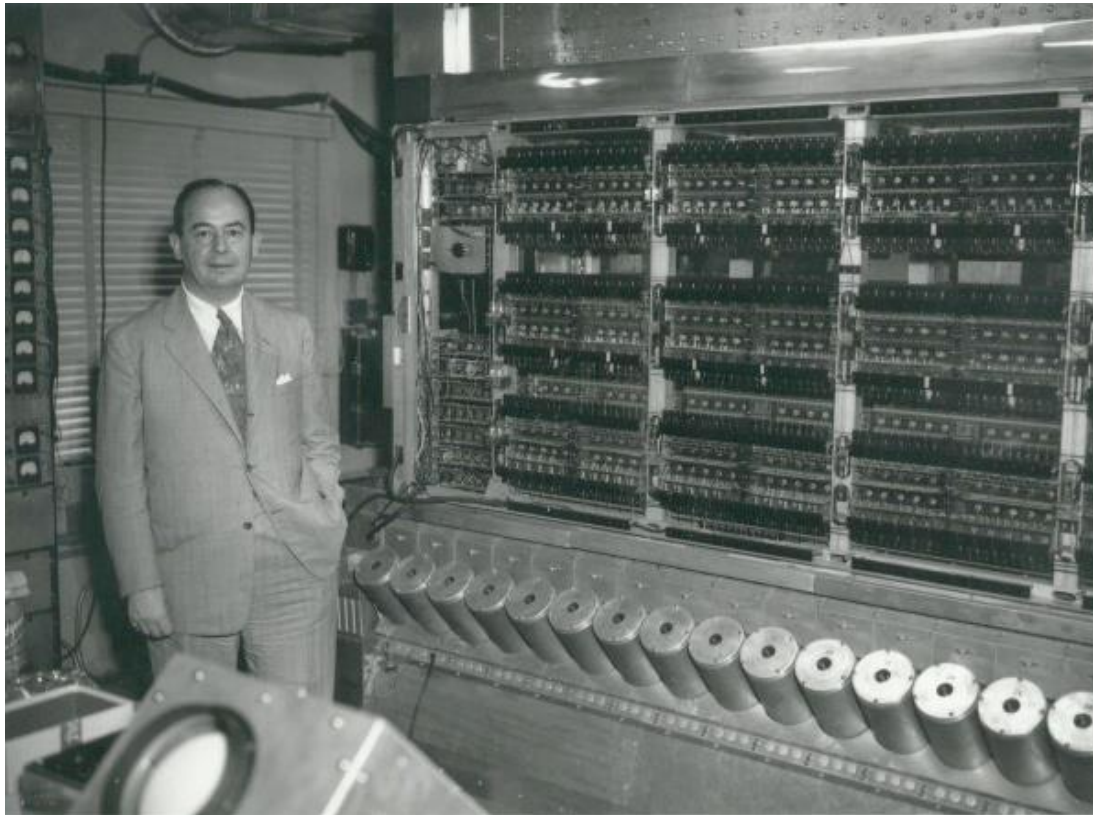
1903 - 1957

1951



□ IAS (Institute for Advanced Studies) Machine

- The first modern computer with stored program (*Von Neumann architecture*)
- Characteristics
 - 450 kg, 1700 vacuum tubes, 140 KW power dissipation
 - Binary computer with 40-bit word and 20-bit instructions



Source: IAS

Simple Batch Systems



❑ **Monitor**

- User submits the job on cards or tape to a computer operator, who batches them together sequentially and places them on an input device
- Monitor is a resident software in main memory
- Monitor reads in jobs one at a time from the input device.
- The current job is placed in the user program area
- The control is passed to the job.
- When the job is completed, it returns control to the monitor
 - User no longer has direct access to the processor

❑ **History**

- The first batch OS was developed by GM in the mid-1950s for use on IBM 701
- By the early 1960s, a number of vendors developed batch OS for their computer systems

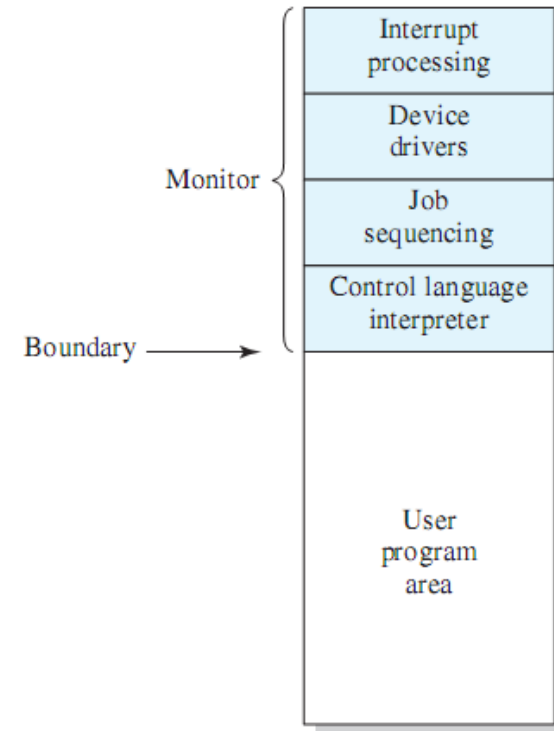


Figure 2.3 Memory Layout for a Resident Monitor

Source: Pearson

Batch Systems: Problems



❑ ***Processor is often idle***

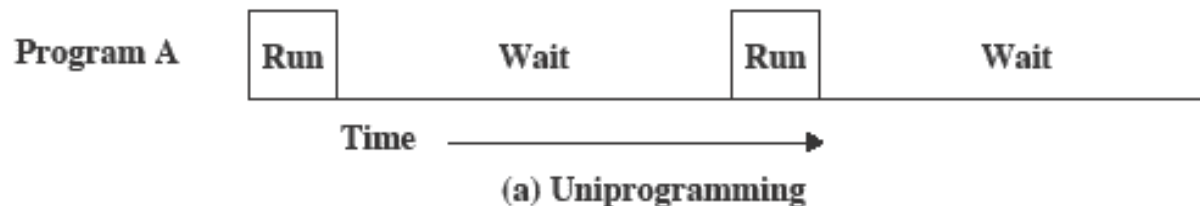
- Even with automatic job sequencing
- I/O devices are slow compared to processor

Read one record from file	15 μ s
Execute 100 instructions	1 μ s
Write one record to file	<u>15 μs</u>
TOTAL	31 μ s

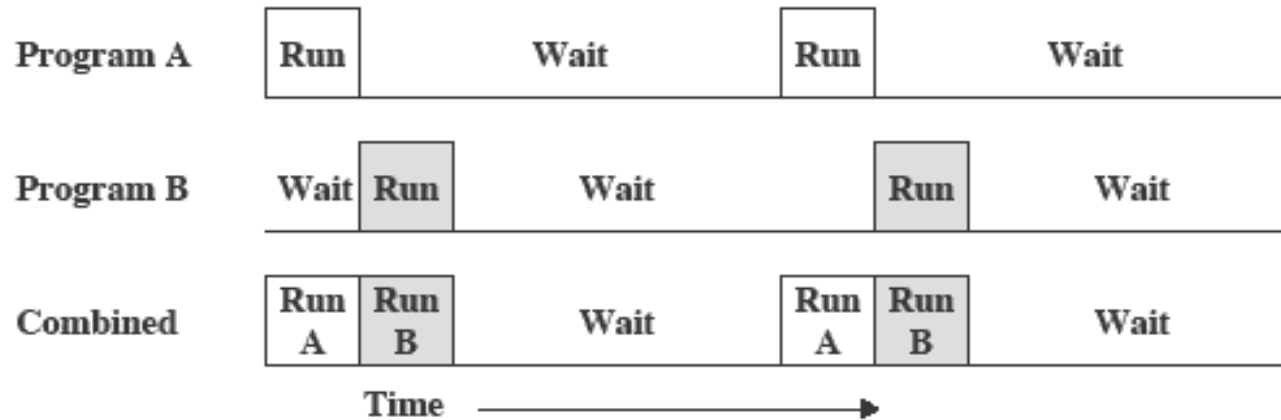
$$\text{Percent CPU Utilization} = \frac{1}{31} = 0.032 = 3.2\%$$

Source: Pearson

Figure 2.4 System Utilization Example



Multiprogrammed Batch System



(b) Multiprogramming with two programs

Source: Pearson

- ❑ ***When one job needs to wait for I/O, the processor can switch to the other job, which is likely not waiting for I/O***

- Also known as multitasking
- Memory can be expanded to hold three, four, or more programs

Multiprogramming Example



Table 2.1 Sample Program Execution Attributes

	JOB1	JOB2	JOB3
Type of job	Heavy compute	Heavy I/O	Heavy I/O
Duration	5 min	15 min	10 min
Memory required	50 M	100 M	75 M
Need disk?	No	No	Yes
Need terminal?	No	Yes	No
Need printer?	No	No	Yes

Source: Pearson

Utilization Histogram

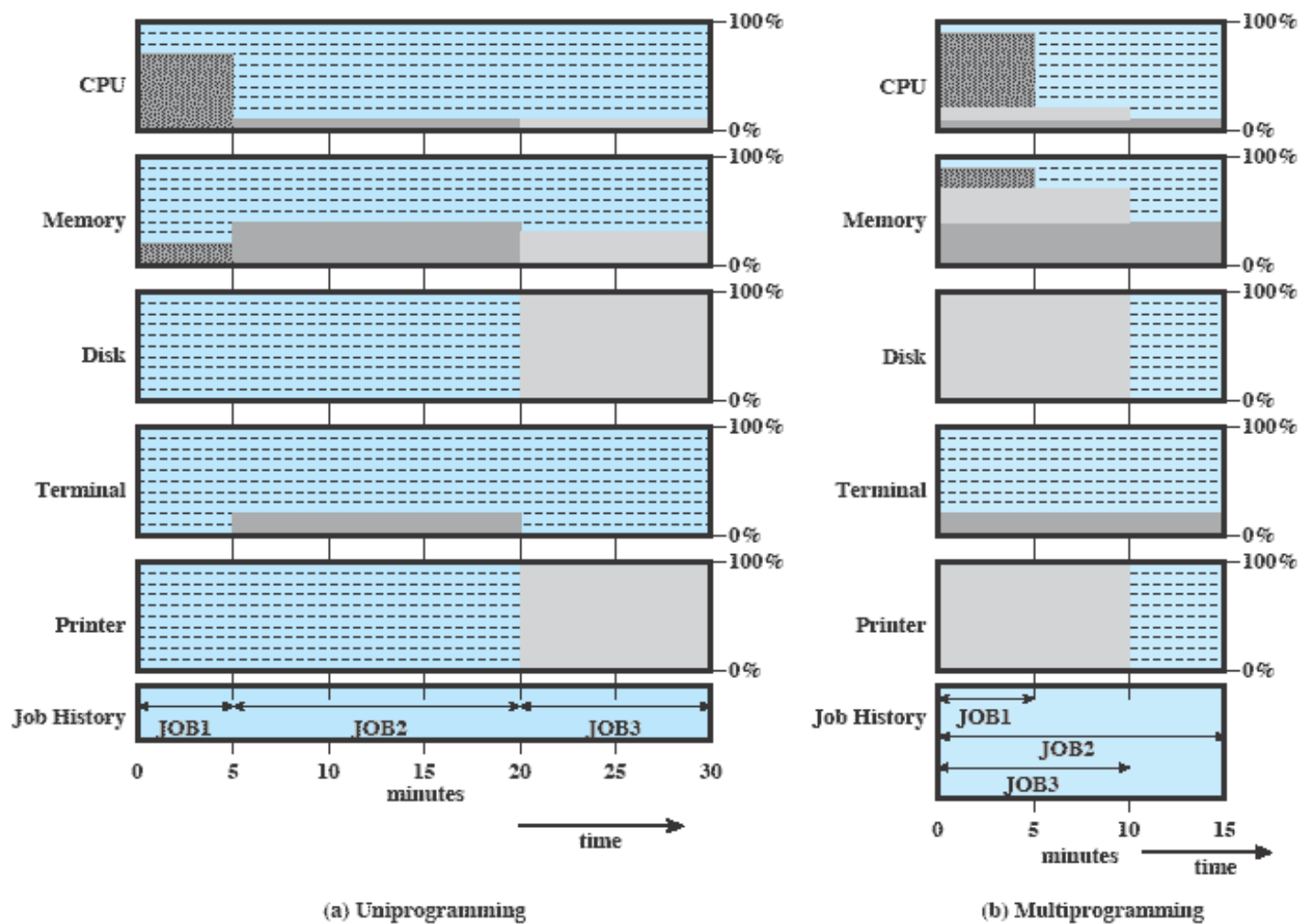


Figure 2.6 Utilization Histograms

Source: Pearson

Effects on Resource Utilization



	Uniprogramming	Multiprogramming
Processor use	20%	40%
Memory use	33%	67%
Disk use	33%	67%
Printer use	33%	67%
Elapsed time	30 min	15 min
Throughput	6 jobs/hr	12 jobs/hr
Mean response time	18 min	10 min

Table 2.2 Effects of Multiprogramming on Resource Utilization

Source: Pearson

Time-Sharing Systems



❑ ***Can be used to handle multiple interactive jobs***

- In a time-sharing system, several users share a computer system through terminals at the same time.
- In this system, *minimizing response time is more important* than maximizing throughput (processor utilization)
- OS interleaves the execution of each user program in *time slice*.
 - Context switching occurs every time slice. The currently running job is suspended and a previously suspended job (or a new job) is resumed (is selected for execution).
- Short jobs or I/O intensive jobs do not have to wait for long compute-intensive jobs

Compatible Time-Sharing Systems



❑ ***CTSS: One of the first time-sharing OS***

- Developed at MIT by a group known as Project MAC
- Ran on a computer with 32,000 36-bit words of main memory, with the resident monitor consuming 5000 words
- To simplify both the monitor and memory management a program was always loaded to start at the location of the 5000th word
 - First developed for IBM 709 in 1961 and later transferred to IBM 7094
 - Running on 7094, CTSS supported 32 users

❑ ***Time Slicing***

- System clock generates interrupts at a rate of approximately one every 0.2 seconds
- At each interrupt OS regained control and could assign processor to another user
- Old user programs and data were written out to disk
- Old user program code and data were restored in main memory when that program was next given a turn

CTSS Operation

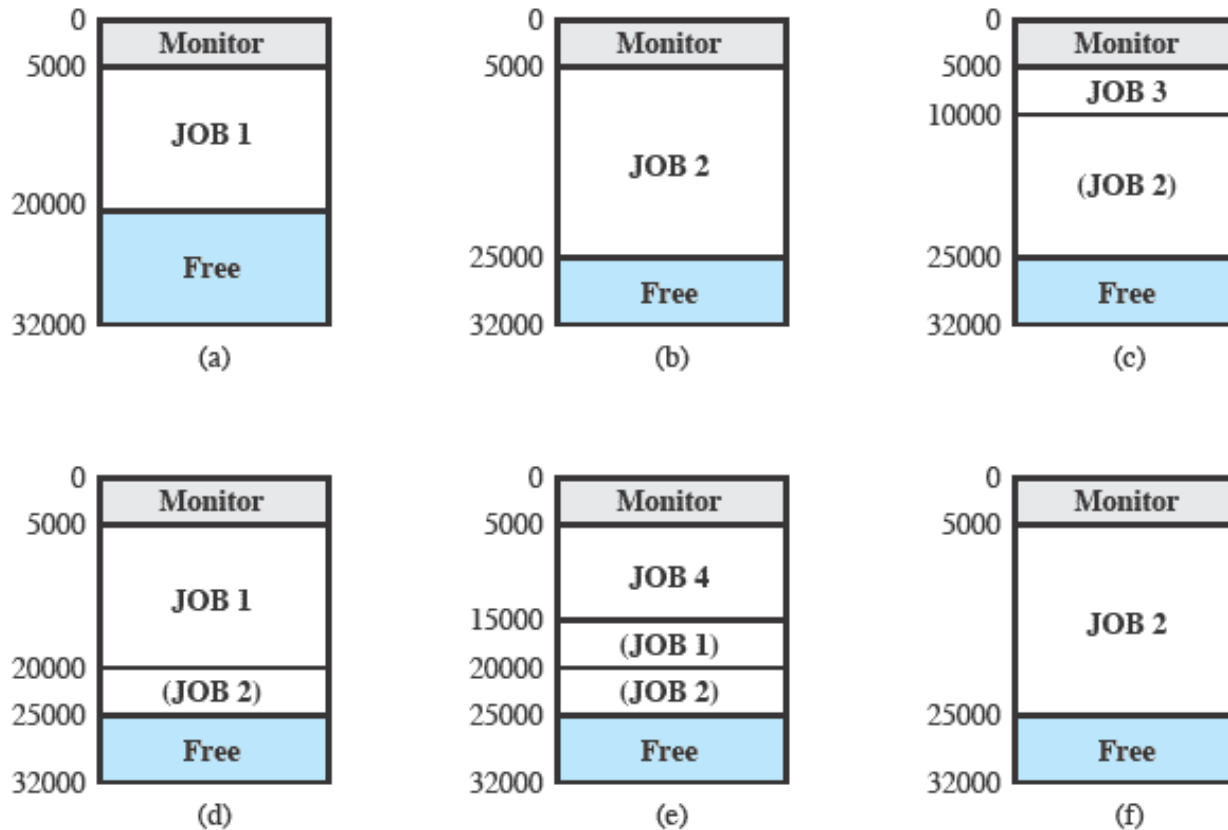


Figure 2.7 CTSS Operation

Source: Pearson

Homework 1



- ☐ ***Read Chapter 1***
- ☐ ***Read Chapter 2***
- ☐ ***Exercise 2.1***
- ☐ ***Exercise 2.2***
- ☐ ***Exercise 2.5***
- ☐ ***Read Chapter 3***